Ask Me Nothing: Learning Need-to-Know Clarification for Financial Agents with Multi-Turn RL

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Anchored in: L3 (Dialogue as a POMDP), L10 (Post-Training), L12 (Evaluation and User Simulation), L15 (Reasoning and Test-Time Compute), L16 (Agents, Tool Use, and Agentic RL).

1 Problem and Why It Is Open

In financial dialogue, clarification requests can collect unnecessary sensitive data, while failing to ask can leave goals underspecified. FINRA Rule 2111 motivates investment-profile checks where applicable (excluding recommendations subject to Regulation Best Interest); Regulation S-P motivates safeguarding customer information (Financial Industry Regulatory Authority, 2012, 2024). These motivate our simplified desk policy, not a claim of regulatory compliance. We ask whether an agent can learn to obtain only task-relevant information while completing a multi-turn task.

Closest prior work. UserBench evaluates preference elicitation in tool-grounded dialogue (Qian et al., 2025); Edwards and Schuster (2026) study clarification in coding agents. CollabLLM learns collaboration through multiturn-aware rewards (Wu et al., 2025). τ-bench evaluates tool-agent-user interaction using policies and final-state checks (Yao et al., 2025). AirGapAgent restricts access to task-relevant data (Bagdasarian et al., 2024), and AgentDAM evaluates data minimization and prompting defenses in web agents (Zharmagambetov et al., 2025). AgentRaft detects cross-tool data over-exposure (Lin et al., 2026); PrivacyLens evaluates contextual privacy norms (Shao et al., 2024a); AgentDojo tests prompt-injection robustness (Debenedetti et al., 2024).

Gap and contribution. CoA already uses GRPO for clarification in visual question answering (Cao et al., 2026). We instead test question and sensitive-data costs jointly in policy-constrained tool dialogue: can they reduce unnecessary collection without suppressing necessary questions? The contribution is this controlled study, not RL for clarification alone.

2 Research Questions

- **RQ1.** Does multi-turn GRPO improve policy-compliant task success over prompting and SFT distillation? *Refuted if* gains vanish against SFT or require more violations.
- **RQ2.** Does pricing unnecessary sensitive requests reduce over-collection and leakage without reducing success? *Refuted if* mandatory items are skipped or failures increase.
- **RQ3.** Are gains explained by fewer redundant questions while retaining necessary ones, rather than a never-ask shortcut? Thinking allocation is exploratory.
- **RQ4.** Do gains transfer to LLM-simulated clients and persist under injected requests for protected data? This is a secondary robustness study.

3 Method and System

BrokerDesk. We will implement a synthetic brokerage desk with profile, position, quote, risk-check, order, and client-query tools, plus outbound counterparty/research messages. A fixed policy specifies position, order-size, restricted-security, and margin constraints. Start with cash-raising and rebalancing; hedging and tax-aware trimming are extensions. Seeded price paths advance between tool turns.

Each information item is labeled *derivable*, *on file*, *mandatory and missing*, *client-only*, or *irrelevant*. Structured field requests make question scoring deterministic; a field already obtained is redundant. Sensitivity tiers are low, sensitive, and restricted. Canary matching measures verbatim disclosure in agent outputs and outbound tool arguments, excluding authorized disclosures under the desk policy. It does not detect paraphrases or

guarantee privacy; incoming observations are not themselves leaks.

Reward and training. For questions n_q, unnecessary field requests U, violations n_v, and unauthorized canary disclosures n_ℓ:

$$R = \mathbf{1}[\text{goal met}] - \lambda_q n_q \\ - \textstyle\sum_{f \in U} \lambda_{\text{tier}(f)} - \lambda_v n_v - \lambda_\ell n_\ell.$$

Repeated requests incur repeated costs; missing mandatory fields count as violations. We will adapt GRPO (Shao et al., 2024b) to groups of complete dialogues from the same task, updating assistant tokens only. LoRA training targets Qwen3-1.7B (Qwen Team, 2025), with 0.6B for smoke tests. A deterministic scripted client answers requested fields from the hidden goal; training progresses from fully specified to underspecified tasks. SFT and SFT+GRPO share the same initial distilled checkpoint to isolate the RL effect.

4 Data, Baselines, and Evaluation

Data. Generate about 300 training, 50 validation, and 100 test tasks with separate clients, seeds, and goal/portfolio combinations. Tune only on validation. All data are synthetic; no human subjects or customer records. Test information labels and policy checks before training.

Baselines. Compare the same model with a privacy-aware prompt, SFT on filtered frontier-model trajectories, a prompted frontier reference, and an AirGap-style minimizer before the prompted small model. All receive the same task information, tool access, and turn limits; report extra minimizer calls and tokens. The frontier reference is not an upper bound.

Metrics. Primary: goal completion with zero policy violations. Also report raw completion, violations, necessary/redundant questions, missed mandatory fields, collection by tier, and unauthorized canary leak rate. Report pass^k for $k = 1, 3$ (success on all k trials, not any one), using five rollouts per test task (Yao et al., 2025). Secondary execution shortfall is signed dollars versus a goal-aware feasible reference on the same price path; report failures separately.

Multi-turn tests. Score the full transcript and final state, with a 12-tool-turn limit. Use three training seeds, paired test tasks, and task-bootstrap 95% confidence intervals plus per-seed scores. Compare scripted clients with a fixed subset using LLM clients (terse, verbose, mistaken, pushy); audit simulator goal drift. Pair clean and injected tool outputs and report attack success. Disaggregate by task and information class; inspect failure transcripts. Core ablations remove privacy cost and question cost separately. Thinking on/off and tool removal are optional diagnostics.

5 Compute, Risks, and Fallback

We plan one A100/H100-class GPU on PACE-ICE, where the team has run course jobs. The unmeasured estimate is 5–10 GPU-hours per 1.7B run: main training and two cost ablations over three seeds total 45–90 GPU-hours, plus SFT and evaluation. Profile a 0.6B run by Nov 15; scale down if memory, runtime, or queue limits intervene. Set an API spending cap before generating distillation data and LLM-client evaluations; log models, tokens, and cost.

Sparse rewards trigger an easier curriculum; never-ask/never-act behavior is checked against tasks requiring missing information and goal completion. Report privacy–success trade-offs, including negative results. If RL fails, retain BrokerDesk and controlled prompted/minimizer comparisons with cost sweeps and robustness tests; make no learned-policy claims.

6 Status and Division of Labor

The repo has interfaces, privacy tiers, canary tests, and a GPU smoke script; the episode loop, generator, checker, and trainers remain to implement. Greyson owns tools and policy; Heeba owns clients, tasks, and privacy; Michael owns training; Canting owns baselines and evaluation. All analyze and write. Targets: minimal environment Nov 9; baseline and first result Nov 19; core runs Dec 2; final paper and code Dec 8. Repository: github.com/IncorvaiaM1/ask-me-nothing.

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